



United International University

Department of CSE

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Report Title: Line Follower Solar Panel Cleaner Robot

Section: A

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Line Follower Solar Panel Cleaner Robot

Introduction: As the global demand for renewable energy sources grows, solar energy has become a crucial part of sustainable power generation. Solar energy has become one of the most promising sources of renewable energy due to its sustainability and low environmental impact. Solar panels are widely used to convert sunlight into electricity, making them an integral part of the global transition to cleaner energy sources. However, a significant challenge faced by solar energy systems is the accumulation of dust, dirt, and debris on the surface of solar panels. This layer of grime reduces the panels' efficiency by blocking sunlight, which in turn limits the amount of energy they can generate.

Traditional methods of cleaning solar panels involve manual labor, which can be both time-consuming and expensive, particularly in large solar farms or areas with harsh environmental conditions (e.g., deserts, rooftops). To address these issues, the need for an automated and efficient cleaning system is crucial.

This project proposes the development of an “Arduino-based Line Follower Solar Panel Cleaner Robot”. The robot will be designed to follow a predefined path or line over the surface of solar panels, using sensors to detect the line and adjust its movement. Equipped with a motorized cleaning mechanism, the robot will autonomously clean the surface of the panels as it moves. This automation will not only reduce the need for manual labor but also ensure that solar panels remain consistently clean, thereby improving their performance and efficiency.



The proposed solution combines simple electronics and robotics with a specific focus on solar panel maintenance. The use of an Arduino Uno microcontroller allows for a cost-effective and easily programmable system, making it accessible for both commercial and residential solar panel installations. By integrating sensors, motors, and a cleaning mechanism, this robot aims to be a reliable and practical solution for maintaining solar panel efficiency in a variety of environmental conditions.

Motivation: The primary motivation for this project arises from the challenges associated with maintaining solar panels, which are critical for converting sunlight into electricity. Solar panels must remain clean and free from dust, dirt, and other debris to function at their full capacity. Over time, natural elements such as dust storms, bird droppings, and pollution can accumulate on the surface of these panels, significantly reducing their ability to absorb sunlight. This results in lower energy production and inefficiencies in the solar energy system.

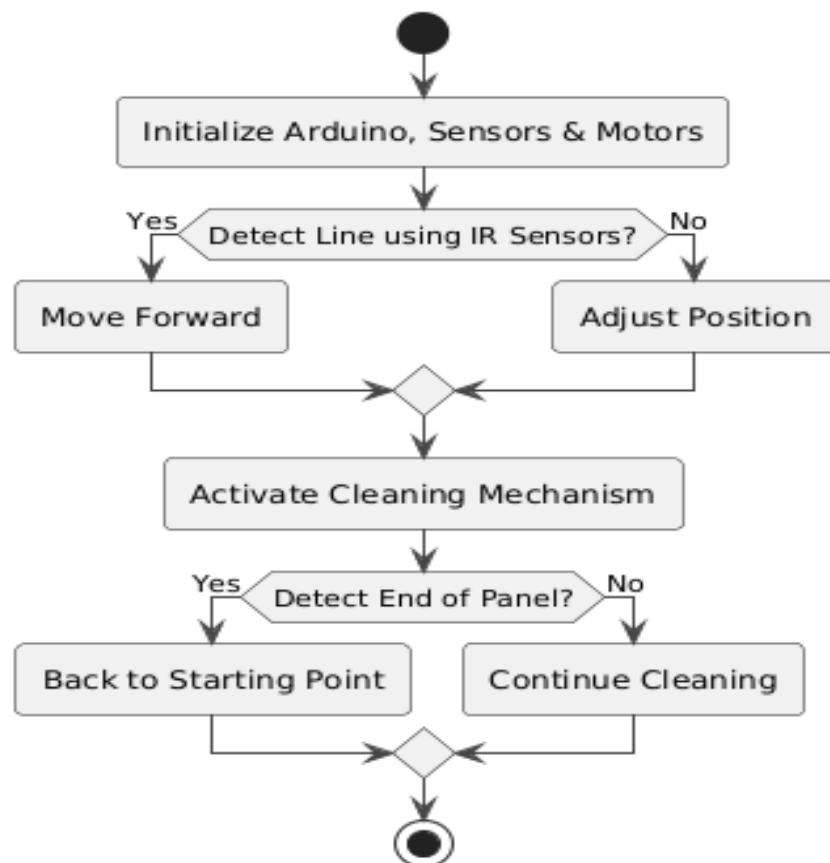
Manual cleaning of solar panels is one solution, but it comes with several drawbacks. First, it can be expensive and labor-intensive, particularly in large-scale solar farms, where thousands of panels need regular cleaning. Hiring personnel to clean these panels regularly can add up to significant costs. They also have potential safety risks for workers who may need to access hard-to-reach areas, such as rooftop solar installations.



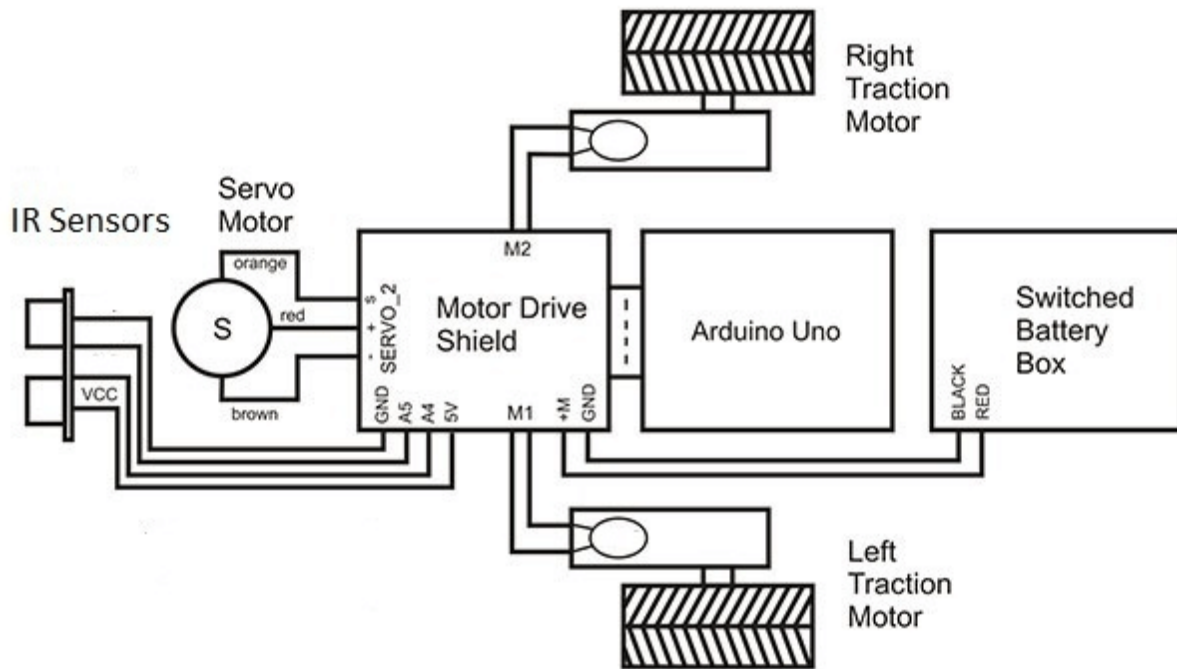
Given these challenges, there is a strong need to reduce human intervention in the cleaning process and adopt a more efficient, automated approach. This is where an Arduino-based robotic system becomes a highly attractive solution. By using line-following technology, the robot can automatically follow a defined path over the solar panels, eliminating the need for human workers to guide or monitor the process. This reduces labor costs, enhances safety, and ensures that the cleaning can be done regularly without depending on human availability or favorable weather conditions.

The project's goal is to develop an affordable, autonomous cleaning robot that can efficiently clean solar panels and maintain their optimal performance. Arduino offers a cost-effective, easy-to-program, and scalable solution. The system can be adjusted to clean panels of different sizes, making it suitable for both small residential setups and large commercial solar farms. By keeping the panels clean, the robot helps maximize the energy output of the solar panels.

Flow Chart: The system flow chart is given below:



System Architecture: The proposed system consists of several components working together to ensure the robot autonomously follows a path and cleans the solar panels. The System architecture is given below:



Components Description: The components of the line follower robot system and its using purpose are given below:

1. Arduino Uno: The Arduino microcontroller will run the program that controls the robot's motors and sensors. It controls the robot's movement by processing inputs from sensors and sending commands to the motors and other components. It is responsible for controlling actions, allowing the robot to clean the solar panels autonomously.

2. Line Follower: The line follower mechanism is used to guide the robot along a predefined path on the surface of the solar panels.

3. Line Following Sensors (Infrared Sensor): These sensors detect the line by differentiating between the black line and the white surface. Based on the line

detection, the sensors send signals to the Arduino, which guides the robot's movement, ensuring it stays on the correct path.

4. Motor Drivers: To control the speed and direction of the DC motors. The Arduino itself can't supply enough current to drive the motors, so a motor driver is necessary.

5. DC Motor: The DC motor will be used to move the robot. The motor driver will control these motors based on sensor inputs and Arduino instructions.

6. Wheel: Wheels attached to the DC motors will allow the robot to move along the line.

7. Chassis for Robot: We will use chassis to mount all the components like the Arduino board, motors, sensors, and battery. It offers the system stability, mobility, and protection to the robot's components.

8. Battery & Battery Holder: The battery powers the robot's motors, sensors, and cleaning mechanism, enabling it to operate autonomously and move across solar panels without external power. The battery holder keeps the battery securely in place, ensuring consistent power delivery and allowing for easy replacement or recharging when needed.

9. Water Pump: The water pump is used for spraying water to assist in cleaning the solar panels. It helps loosen stubborn grime, making the cleaning mechanism more effective in cleaning the solar panels.

10. DC Switch: We use a switch to turn the robot on and off easily.

11. Cleaning Mechanism: We will use water, cleaning spray and foam to clean the solar panel.

USB Cable: The USB cable is used for programming and powering the Arduino during development and testing.

Code Instruction: The code instructions control robot movement and cleaning actions. It processes inputs from sensors and decides how the robot should respond, ensuring it stays on track and cleans efficiently.

Solar Panel: We will need solar panels because it will be needed to clean.

Jumper Wires: Jumper wires connect circuit components and ensure electrical connections.

Gantt Chart (Weekly Plan): We complete our project within 3 weeks. The work flow is given below:

Week	Task	Duration (Days) & work (%)
Week 1	1. System Design & Component Selection: Arduino, sensors, motors, cleaning mechanism, etc. 2. Purchase Components & Basic Circuit Assembly 3. Mechanical Assembly: Attach Motors, Wheels, Sensor	=> 7 Days => 25% will complete
Week 2	1. Integrate Line-Following Sensors with Motor Control 2. Program Line Following Logic & Test Sensors 3. System testing and debugging	=> 7 Days => 60% will complete
Week 3	1. Implement Cleaning Mechanism 1. System Integration: Combine Line Following & Cleaning Mechanism 2. Full System Testing & Debugging 3. Final Adjustments & Documentation	7 Days => 100% will complete

Budget Estimation: The predictable Budget estimation is given below:

Component	Quantity	Cost
1. Arduino Uno	1	530
2. Line Follower	1	100
3. IR Sensors (Line Following)	2	200
4. DC Motors	2	275+275
5. Motor Driver (L298N)	1	210
6. Cleaning pipe & Foam	1	150
7. Wheels/Chassis	1 set	300
8. Battery	4	250+250+150+150
9. Battery Holder	3	80+30+30
10. Water Pump	1	50

11. DC Switch	3	20+10+10
12. USB Cable	1	135
13. Cleaning Spray	1	120
14. Solar Panel	1	200
15. Miscellaneous (wires, connectors, etc.)		250
16. Other Cost		500
Total Cost		4275

Conclusion: The Arduino-based Line Follower Solar Panel Cleaner Robot is a cost-effective and autonomous solution to maintaining solar panel cleanliness. By automating the cleaning process, this robot can increase solar panel efficiency and reduce maintenance costs. This reduces labor costs, improves safety, and ensures that solar panels maintain their maximum energy efficiency by staying free from dust and debris. The system will be developed using affordable components, ensuring scalability and adaptability to various sizes of solar installations. The robot's scalable design makes it suitable for both small residential installations and large commercial solar farms, promoting the long-term sustainability and performance of solar energy systems. We aim to complete this project within three weeks, according to the proposed schedule.

Thank You